

Mohammad Abrar R

abrarnoobdev@gmail.com | github.com/rarba-noobdev | linkedin.com/in/mohammad-abrar-462697286 | Chennai, India

EDUCATION

B.Tech — Artificial Intelligence & Data Science

2023–2027

B.S. Abdur Rahman Crescent Institute of Science and Technology, Chennai

CGPA: 7.23

PROJECTS

SentinelAuth AI

IOB Cybernova 2026 · Team Tech Titans

Android · TensorFlow Lite · MediaPipe · MiniFASNetV2 · Python · TFLite

- Designed a fully on-device, three-stage deepfake detection pipeline for Android: MediaPipe face detection (Stage 1), MiniFASNetV2 passive anti-spoofing PAD (Stage 2), and a custom `deepfake_v1.tflite` binary classifier (Stage 3, 7.8 MB, 192×192, AUC 0.8772)
- Eliminated cloud dependency for banking-grade firewall compatibility; resolved PAD class-index bug (index 1 vs 2) and validated normalization flow for consistent spoof rejection
- Researched and benchmarked mobile-deployable architectures: InspireFace Android SDK, facenox MiniFASNetV2-SE INT8 (Apache 2.0), and NPR (CVPR 2024) as a quantization-robust Stage 3 candidate

Yoho — Campus Social Hub

React · TypeScript · Vite · Tailwind CSS · Supabase Realtime · Razorpay · Resend

- Built a Gen-Z campus social platform competing with Unstop, featuring native event chat (Supabase Realtime), a gamified trophy case (XP, badges, streaks), and a bento-grid trending feed
- Integrated Razorpay for paid event ticketing, Resend for transactional emails; diagnosed and fixed RLS infinite recursion bug on the profiles table
- Conducted a full SEO audit (14/100 baseline), initiated structured-data and sitemap implementation to address zero Google index presence from client-side SPA routing

LapKart — Laptop Components Quick Commerce

Solo Founder · Ongoing

React · Vite · TanStack Router · TanStack Query · Supabase · Razorpay · Shiprocket · Ola Maps

- Building a quick-commerce platform for laptop parts (RAM, SSD, batteries, displays) targeting 2–4 hour delivery in Chennai, with dark-store fulfilment architecture and live rider tracking via Ola Maps SDK
- Integrated Shiprocket for logistics orchestration and Razorpay for checkout; identified SP Road / Ritchie Street as primary supplier network

DWSI IT Asset Manager — Dongwoo Surfacetech India

Client Delivery

React · TypeScript · Tailwind · shadcn/ui · Supabase · AG Grid Community · SheetJS · Zustand

- Delivered a production asset-tracking system managing 93 assets across 20 departments and 2 divisions (HT, FMD) for a manufacturing client
- Built Excel bulk-upload via position-based column mapping (bypassing duplicate-header limitations of name-based parsing) using SheetJS; AG Grid Community for high-performance tabular views with filtering, sorting, and CSV export

Cognitive Echo Sentinel — Neurological Condition Detection via Voice

Python · scikit-learn · XGBoost · Librosa · TQWT · StratifiedGroupKFold

- Engineered 200+ acoustic biomarkers (jitter, shimmer, RPDE, DFA, PPE, HNR, MFCC, TQWT sub-bands) to train an XGBoost/Random Forest/SVM ensemble for voice-based Parkinson's severity detection
- Corrected PPE formula bug, replaced standard scaling with QuantileTransformer to resolve saturation, and applied StratifiedGroupKFold to prevent subject-level data leakage across folds

SolarSaathi — AI-Powered Rooftop Solar Advisor

Planet-X Hackathon 2026 · JCE Chennai · SDG 7

React 18 · TypeScript · Claude API · Google Earth API

- Built a 5-step wizard that analyses rooftop area via Google Earth (zero-cost measurement), applies TANGEDCO telescopic slab tariff calculations and sanctioned load capping, and generates personalised solar ROI projections aligned with PM Surya Ghar targets
- Integrated Claude API for conversational advisory and dynamic system-sizing recommendations for households and SMBs

OpenClaw — Personal Always-On AI Agent

Azure VM (B1ms) · Claude API (Haiku 4.5 / Sonnet) · Node.js · WhatsApp · Gmail · Google Calendar

- Self-hosted AI agent running on Azure B1ms, accessible via WhatsApp with Claude Haiku 4.5 as primary model and Sonnet as fallback; equipped with `gogcli` for Gmail/Calendar/Drive operations and browser automation skill
- Implemented a self-improving-agent skill loop with persistent config and a roadmap for Sarvam AI Tamil/Hindi voice integration via Twilio

FORGE — Gaming PC Builder Design System

Portfolio / Creative

React · TypeScript · Vite · Tailwind CSS · Framer Motion

- Designed and built a cyberpunk-aesthetic PC configurator with six unique component card types (CPU, GPU, RAM, Storage, PSU, Cooling), live wattage and bottleneck calculation, and a glassmorphism design system
- Implemented deep ocean blue / cyber crimson token palette, animated stat rings, type-specific GPU fan spin and RGB RAM sweep animations, and a full component library reference (buttons, badges, filters, modals)

BQ728 & Buck IC Interactive Circuit Animations

Educational / Hardware

HTML · CSS · Vanilla JS · SVG

- Built two interactive 18-step and 10-step signal-flow animations for laptop motherboard circuits (BQ728 charging IC and dual-channel 3.3V/5V Buck IC) to visualise real-time MOSFET switching, CSR sensing, PHASE lines, VFB feedback, and bootstrap voltage
- Implemented cumulative signal highlight, step-by-step progression with auto-play timer, and anatomically correct MOSFET gate terminal orientation for use as hardware-repair learning tools

TECHNICAL SKILLS

Languages	Python, TypeScript, JavaScript, C
Frontend & UX	React, Vite, SvelteKit, Tailwind CSS, shadcn/ui, Framer Motion, Radix UI, Figma, Google Stitch, Glassmorphism, Design Systems
App Development	Android (Java), TFLite on-device inference, Capacitor (Web→Android APK)
Backend & BaaS	Express, FastAPI, Supabase, PostgreSQL, pgvector, Firebase
ML / AI	TensorFlow Lite, scikit-learn, XGBoost, MediaPipe, HuggingFace Transformers
Electronics	Laptop board-level repair, MOSFET testing & diagnosis, SMD component testing, circuit fault isolation, Buck/LDO converter analysis, BQ-series charging ICs
Embedded Systems	8051 microcontroller, Keil µVision, RTOS (Rate Monotonic Scheduling, EDF), multimeter diagnostics
Tools & Platforms	Git, GitHub, VS Code, Android Studio, Claude Code, TI-84 Plus, Google Colab, Linux

ACHIEVEMENTS

- **1st Place** — TechFusion, Crescent Institute of Science & Technology
- **1st Place** — Project Expo, Crescent Institute of Science & Technology
- **Finalist** — IOB Cybernova 2026 National Hackathon, SentinelAuth AI (Team Tech Titans)
- **Shortlisted** — Google Solution Challenge 2026, AI-01 Emergency Incident Fusion System
- **Participant** — Planet-X Hackathon 2026, JCE Chennai (SolarSaathi, SDG 7)

CERTIFICATIONS

- Responsive Web Design — **freeCodeCamp**
- JavaScript Algorithms and Data Structures — **freeCodeCamp**
- Front End Development Libraries — **freeCodeCamp**
- Generative AI Fundamentals — **Google Cloud Skills Boost**
- Introduction to Machine Learning — **NPTEL**